How Progress Ends
by Carl Benedikt Frey

European Economic History — Spring 2026

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April 2026

University of California, Berkeley
with Barry Eichengreen

Roadmap

1. About the author — is a book the right venue?
2. The central thesis
3. China's reversal of fortune
4. Europe's decentralized advantage
5. Paths to industrialization: Britain, Germany, Russia, Japan, America
6. Cold War divergence and modern stagnation
7. Critical assessment
8. Discussion

Lightning Round

In 30 seconds or less:

What did you think of the book?

About the Author: Carl Benedikt Frey

- Dieter Schwarz Associate Professor, Oxford Internet Institute
- Director of the Future of Work programme, Oxford Martin School
- Previously: *The Technology Trap: Capital, Labor, and Power in the Age of Automation* (2019)
- Research focus: economics of AI, technological change, and labor markets
- Ph.D. in Economic History
- Frequent contributor to policy debates on automation and inequality

Oxford University Press, February 2025 — 378 pages

Is a Book the Right Venue?

What the format enables:

- Sweeping narrative across 2,000+ years
- Synthesizes growth theory, political economy, history of science
- Accessible beyond academia

What it sacrifices:

- No formal model — mechanism never precisely defined
- Evidence is illustrative, not causal
- Hard to falsify: which cases would *disprove* the thesis?
- Selection of episodes may confirm the prior

Could the core claims have been a series of papers instead?

The Central Thesis

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Societies that centralize too much stifle the exploration needed for breakthrough innovation. Progress can — and has — ended before.

The Framework: Exploration vs. Exploitation

Exploration
(Decentralized)

- Trial and error
- Autonomous agents
- Frontier innovation
- High variance, high upside
- Markets, federalism, universities

Exploitation
(Centralized)

- Scaling known solutions
- Top-down coordination
- Catch-up growth
- Efficient but rigid
- Bureaucracies, state planning

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Key insight: Countries at the technological **frontier** need exploration; countries **behind** the frontier can grow via exploitation — but only temporarily.

Contrast: Greif, Mokyr & Tabellini, Two Paths to Prosperity

Shared ground:

- Both compare Europe and China over the long run (1000–2000)
- Both reject simple "Europe was better" narratives
- Both emphasize political fragmentation as key to Europe's divergence

Where they diverge:

- **GMT:** Europe's fragmentation → warfare → fiscal-military states → capital-intensive growth. China's unity → peace → labor-intensive, Smithian growth. *Two viable paths*, not one failure.
- **Frey:** Fragmentation → innovation. Centralization → stagnation. More normatively loaded — one path ultimately wins at the frontier.

Does Frey's framework undervalue the Smithian growth path?

The Needham Puzzle: Why Did China Fall Behind?

China's early lead:

- Bureaucratic state from 221 BC (Qin unification)
- Printing, gunpowder, compass, paper — centuries before Europe
- Civil service examinations from 605 AD
- Song dynasty (960–1279): "an industrial revolution that almost was"

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What went wrong?

- Political unity eliminated interstate competition
- The exam system channeled talent into bureaucracy, not innovation
- Ming sea ban (1371): shut down maritime exploration
- Patronage economy under Qing: loyalty $>$ innovation

China: The Fiscal Capacity Trap

Frey's key evidence:

- Tax revenue per capita *declined* from Song to Qing
- By 1900, China's fiscal capacity was a fraction of Europe's
- A vast empire with a surprisingly weak state

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The paradox:

- Centralized enough to *block* innovation from below
- Too weak to *invest* in modernization from above

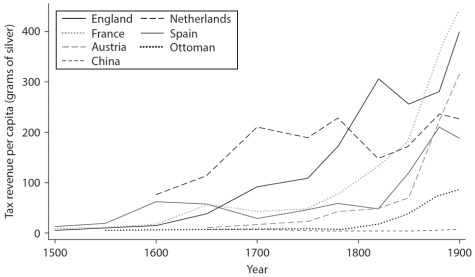

FIGURE 2.3. Comparative fiscal state capacity around the world, 1500–1900
Source: Karaman and Pamuk, "Different Paths to the Modern State in Europe."
For China: Brandt, et al., "From Divergence to Convergence," 45–123, Table 3.

The Rise of Europe: Decentralization as Engine

Why Europe? The fall of Rome *enabled* progress.

- **Political fragmentation:** Hundreds of competing polities
- **Germanic traditions:** Assemblies, consensual governance
- **Parliaments:** León (1188), Magna Carta (1215) — constraints on rulers
- **Communes:** Self-governing cities (Italian city-states, Hanseatic League)
- **The Church:** Banned cousin marriage $\rightarrow$ weakened clans $\rightarrow$ strengthened individualism and voluntary associations

Key mechanism
Interstate competition + exit options for talent $\rightarrow$ rulers had to offer good institutions to attract productive people.

Britain: The Model of Decentralized Innovation

- Glorious Revolution (1688): parliamentary sovereignty, property rights
- Patent system: credible commitment to protect inventors
- No guilds monopolizing knowledge; artisan-driven experimentation
- Industrial Revolution as *bottom-up* phenomenon

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Frey's interpretation:

- Britain succeeded not because of a strong state directing innovation...
- ...but because the state *could not* easily block innovators
- Parliament represented merchant/industrial interests
- Creative destruction was politically tolerated

Germany/Prussia: Bureaucratic State Capitalism

- Frederick the Great's bureaucratic reforms: meritocratic, efficient
- **Education first:** Prussia built mass schooling *before* industrialization — the reverse of Britain, where industry drove education demand
- State-directed industrialization: railroads, steel, chemicals
- **Universities:** Humboldt model — research universities with genuine autonomy
- German universities dominated Nobel Prizes until 1933

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The catch:

- Worked brilliantly for *catch-up* (replicating British innovations)
- State coordination scaled applied science effectively
- But: when regime changed to Nazi control, university autonomy collapsed
- Brain drain (Einstein, Fermi, von Neumann...) $\rightarrow$ US inherited the frontier

Russia and Japan: Imitation Without Exploration

Russia/USSR

- Weak state *and* weak society
- Serfdom until 1861
- Central planning as "exploitation machine"
- Impressive catch-up (steel, space)
- Virtually zero breakthrough inventions

Japan

- Meiji state-led modernization
- Pilot factories $\rightarrow$ privatized to zaibatsu
- Labor repression funded capital accumulation
- Caught up fast, but breakthrough inventions: negligible

"To paraphrase one scholar, if Japan's experience teaches any single lesson, it is that myriads of simple improvements to established technologies can take a country a long way."

America: The Innovation Powerhouse

Why the US took the lead:

- **Federalism:** States as "laboratories of democracy" (Brandeis)
- **Constitution:** Patent clause enshrined innovation incentives from 1787
- **Democratic innovation:** Broad-based patenting, not just elites
- **Frontier individualism:** Geographic mobility, weak guilds
- **ARPA model:** Decentralized funding, high autonomy for researchers

Key contrast with Europe:

- US patent holders were far more socially diverse
- Innovation was not a preserve of the educated elite
- "Democratized" invention $\rightarrow$ more experimentation

ARPA and the Internet: Decentralized Innovation in Practice

The ARPA model (est. 1958):

- Small agency, flat hierarchy, program managers with high autonomy
- Funded high-risk research with no immediate commercial application
- Trusted researchers to define problems, not just solve assigned ones

The Internet as case study:

- ARPANET (1969): designed for resilience, not central control
- Open protocols (TCP/IP) enabled permissionless innovation
- No single firm or government directed the web's development
- Result: the most consequential innovation platform of the 20th century

Frey's strongest example: state funding + decentralized execution = breakthrough.

Cold War: The Great Experiment

USSR vs. China after 1949: Two paths of communism

Soviet Union

- Rigid central planning
- *Nomenklatura* system: political loyalty over competence
- Khrushchev's Sovnarkhoz reform failed
- Stagnation by 1970s

China (post-1978)

- Deng's *gradual* reforms
- Regional decentralization
- Special Economic Zones (Shenzhen)
- "Crossing the river by feeling the stones"

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Frey's reading: China's success was *decentralization within an autocracy* — but this has limits...

[Fig. 10.2: GDP per capita, USSR vs. China, 1920–2010]

The Dictator's Dilemma

Autocrats **need** innovation for legitimacy and power,
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- Innovation requires free information flows — but these threaten regime control
- Autonomous researchers may produce politically inconvenient truths
- Successful entrepreneurs become independent power centers
- Xi Jinping's recentralization: crackdowns on tech firms, universities, SEZs

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Historical parallel
Ming China, Nazi Germany, late USSR — all centralized at the cost of innovation.

Modern Stagnation: Not Just China

Frey argues the US faces its own innovation slowdown:

- Declining research productivity (more researchers, fewer breakthroughs)
- University bureaucratization: compliance, administration, risk aversion
- Regulatory accumulation: harder to experiment (pharma, energy, construction)
- Geographic sorting: innovation concentrated in fewer places
- Political polarization undermining institutional trust

The "veneer of progress": We may be coasting on past institutional investments while the foundations erode.

Epilogue: AI and Moravec's Paradox

Moravec's Paradox:

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Frey's provocation:

- AI is extremely good at *exploitation* — pattern recognition, optimization
- But genuine *exploration* — creative leaps, paradigm shifts — remains elusive
- AI may *accelerate* the centralization-decentralization tension
- States with AI can monitor and control more effectively...
- ...but breakthrough innovation still requires decentralized human creativity

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Will AI make the dictator's dilemma easier or harder to resolve?

Critical Assessment: Strengths

- **Ambitious scope:** 2,000+ years of history in a coherent framework
- **Clear thesis:** Centralization–decentralization as an organizing principle
- **Rich historical detail:** Moves fluently between China, Europe, US, Japan, Russia
- **Policy relevance:** Speaks directly to current debates on China, tech regulation, AI governance
- **Bridges literatures:** Connects Acemoglu & Robinson, North & Weingast, Hayek, Schumpeter, and modern growth theory

Critical Assessment: Potential Weaknesses

1. **Selection on the dependent variable?**
   We observe cases where centralization $\rightarrow$ stagnation. But what about successful centralized innovation? (DARPA, Manhattan Project, Apollo, mRNA vaccines?)
2. **Vague mechanism:**
   How much decentralization is optimal? The framework is qualitative — hard to falsify or test precisely.
3. **Europe's fragmentation was also violent:**
   Centuries of warfare, colonialism, and destruction. Was fragmentation a net positive, or did it come with enormous costs Frey underweights?
4. **Geography and contingency:**
   How much is explained by institutions vs. geography, disease, demography? (Diamond, Jared; Galor, Oded)
5. **The US story is incomplete:**
   Slavery, land dispossession, and exclusion were central to American development — largely absent here.

Critical Assessment: Potential Weaknesses (cont.)

6. **Catch-up vs. frontier distinction:**
   Is this binary too clean? South Korea, Taiwan, and Israel moved from catch-up to frontier with significant state involvement.

7. **What about small open economies?**
   The book focuses on great powers. How does the framework apply to the Nordics, Switzerland, Singapore?

8. **Defining "progress":**
   Frey focuses on technological invention. But progress also includes institutions, welfare, equality, sustainability. Are these always correlated?

Key Takeaways

1. **Progress is fragile:** History shows repeated cases of technological stagnation following centralization
2. **Catch-up $\neq$ frontier growth:** Different institutional arrangements serve different stages of development
3. **The dictator's dilemma is real:** Autocracies face an inherent tension between control and innovation
4. **Complacency is the risk:** Even democracies can drift toward institutional sclerosis

"The economic organization that excels at inventing the industrial future is not necessarily the most suitable for catching up to an existing target." (p. 13)

Thank you!

Thank you for listening!

Questions?